Sergio Cordero Calvimontes

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Summary

Mechatronics Engineer with *stricto sensu* master's degrees in Mechanical Engineering and Computational Modelling. Experience in numerical simulation, scientific computing and the development of computational tools. Main focus on multiscale modelling, composite materials and thermomechanical processes. Secondary focus on teaching electrical and electronics subjects.

Education and Projects

EEIMVR-MCCT, UFF — Rio de Janeiro, Brazil
Master's degree in Computational Modelling in Science and Technology — 2022–2024

- Computational homogenisation procedures for multiscale analysis.

EEIMVR-PGMEC, UFF — Rio de Janeiro, Brazil
Master's degree in Mechanical Engineering — 2016–2018

- Numerical simulation of electrothermal-mechanical processes in welding.

UPB; revalidation: EESC / USP — Cochabamba, Bolivia / São Paulo, Brazil
Bachelor's degree in Mechatronics Engineering — 2008–2014

- Indirect measurement of cutting forces in CNC processes.

Universidade Estácio de Sá — Angra dos Reis, Brazil
Bachelor's programme in Computer Science, distance learning — 2025–present

FORMA BRASIL EDUCACIONAL, CNE / CEB — Paraíba, Brazil
Technical Diploma in Mechatronics — 2024

Career Outlook

Technical and scientific research: Consolidation of manuscripts and supporting material for the submission of articles on computing, materials modelling and numerical simulation.

Academic development: Seeking a doctoral opportunity, with a technical and scientific portfolio organised for applications.

Technical activity: Sitting public-service examinations as a strategy for continuous updating. Founder of Editora Argus Kepheus, a publisher focused on technical and scientific publications.

Selected Publications

Calvimontes, S. C. **Projeto elétrico industrial básico: fundamentos, dimensionamento e documentação técnica.** 1st ed. e-book. Angra dos Reis: Editora Argus Kepheus, 2026. ISBN 978-65-976344-0-8.

Calvimontes, S. C. *et al.* **Computational homogenization of linear viscoelastic multi-phase composites: Effective relaxation estimation in perovskite-based solar panels.** *Mechanics of Advanced Materials and Structures*, v. 33, n. 1, article 2456105, 2026. [DOI]

Rodríguez-Ramos, R.; Rodríguez-Bermúdez, P.; Calvimontes, S. C.; *et al.* **Computational homogenization for the estimation of overall properties in linear viscoelastic composites.** *Archive of Applied Mechanics*, v. 95, n. 8, article 191, 2025. [DOI]

CV Lattes

Skills

Languages: Spanish, native; Portuguese, fluent, Celpe-Bras: 7+; English, advanced, IELTS: 7+.

Programming: `MATLAB` / `GNU-Octave`, `Python`, `C++`, `Java`, `Julia`; *web development*: `HTML`, `CSS`.

Tools: LaTeX / TeXstudio, Git / GitHub, MS Office, Inkscape, QElectroTech.

Additional Training

Electrical and electronics: electrical installations: NBR 5410, NBR 5413, NBR 5419 and NBR 14039; electrical drive and interlocking circuits; electrical and lighting documentation: QElectroTech; DIALux / Relux.

Mechanics: CAD / CAM / CAE: SolidWorks / OnShape / FreeCAD, including BIM features and integration; AutoCAD / ZWCAD / BricsCAD; MasterCAM / Vectric Aspire / VCarve; DEFORM-3D; ANSYS; Linkage; G / M codes for CNC; prototyping and industrial technical drawing with additive manufacturing in ABS and other polymeric materials; failure analysis and strength of materials; metallography.

Automation: Ladder Logic for PLCs; selection of sensors and transducers; signal acquisition and conditioning; measurement of electrical and physical quantities in automated systems.

Computing: C; PIC, Arduino and ESP32 microcontrollers; prediction of steel properties with AI.

Generative AI: prompt engineering; LLM-related tools, such as agents and tokenisers.